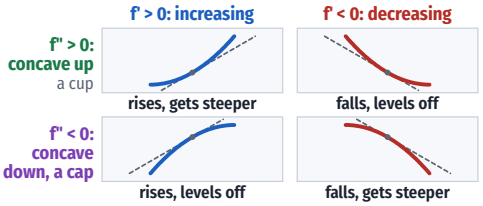
1 The four shapes from the signs of f' and f''



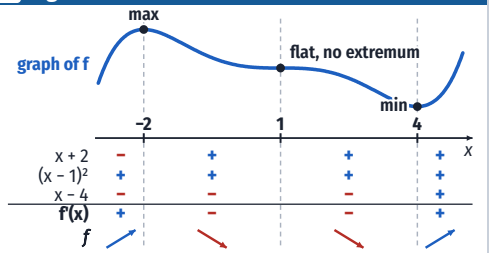
Dashed: the tangent line. Cups lie above it and caps lie below it.

Label: each interval as increasing or decreasing, and as concave up or concave down.

1. The sign of f' picks the column and the sign of f'' picks the row. Join the pieces at the special points.

- A relative max is the top of a cap, a relative min is the bottom of a cup, and an inflection point is where the bend flips.
- On a graph of f': f rises where f' is above the x-axis, and f is concave up where f' is rising.

2 Sign chart on a factored f'



Label: the break numbers, one row per factor, the row for f', and an arrow for f in each interval.

1. The break numbers are the zeros of each factor and the numbers not in the domain.
2. Test one value inside each interval, write the sign of each factor, and multiply down the column.

- Check one column with numbers: f'(0) = (2)(1)(−4) = −8, a minus.
- $(x - 1)^2$ is never negative, so f' keeps its sign at 1 and there is no extremum there.

3 First derivative test: max, min, or neither



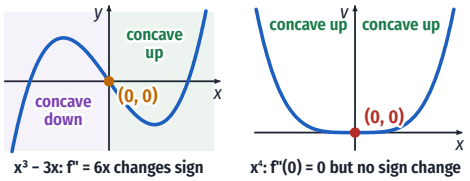
relative max	relative min	no extremum	min at a cusp
f': + then −	f': − then +	f': + then +	f'(0) does not exist

Label: c, the sign of f' on each side, and the point (c, f(c)).

1. A critical number is a c in the domain where f'(c) = 0 or f'(c) does not exist.
2. Read the sign of f' on each side of c, and compute f(c) from f itself.

- Plus then minus is a relative max. Minus then plus is a relative min. No sign change means no extremum.
- $f(x) = x^{2/3}$ has $f'(x) = 2 \div (3x^{1/3})$, which does not exist at 0, so 0 is a critical number.

4 Concavity and inflection points



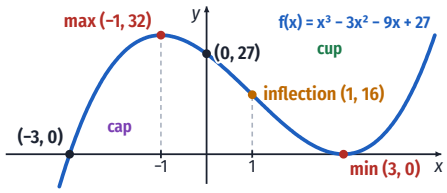
$x^3 - 3x$: f'' = 6x changes sign — x^4: f''(0) = 0 but no sign change

Label: the concave up and concave down intervals, and each inflection point (c, f(c)).

1. Factor f'' and make its sign chart. Plus means concave up and minus means concave down.
2. An inflection point needs f'' to change sign at a c where f is continuous.

- x^4 has f''(0) = 0, but f'' > 0 on both sides, so (0, 0) is the bottom of a cup and not an inflection point.
- A sign change of f'' at a vertical asymptote is not an inflection point: the graph has no point there.

5 Curve sketching checklist, one full example

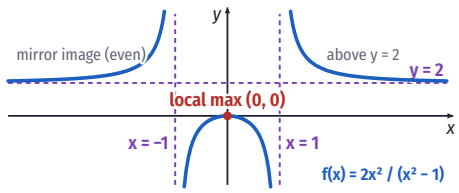


Label: the intercepts, each extremum and inflection point with coordinates, and dashed asymptotes.

1. Find the domain, intercepts, symmetry, and asymptotes. Here $f = (x - 3)^2(x + 3)$, so it crosses at −3 and touches at 3.
2. $f' = 3(x - 3)(x + 1)$ gives the max (−1, 32) and the min (3, 0).
3. $f'' = 6(x - 1)$ gives the inflection point (1, 16).
4. Fill in the table, plot the points, and join them with the shapes from card 1.

Interval	(−∞, −1)	(−1, 1)	(1, 3)	(3, ∞)
f' and f''	+ and −	− and −	− and +	+ and +
Shape	rising cap	falling cap	falling cup	rising cup

6 Asymptotes of a rational function

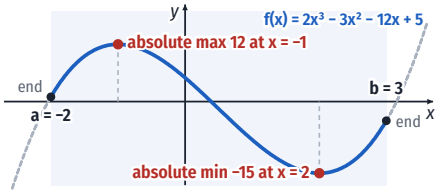


Label: the dashed lines x = −1, x = 1, and y = 2 with their equations, and the local max (0, 0).

1. Cancel common factors. Each zero of the denominator that is left gives a vertical asymptote. A cancelled factor gives a hole.
2. Near x = 1 the top is near 2 and the bottom is tiny, positive on the right and negative on the left.
3. Divide the top and the bottom by x^2. The limit at both ends is 2.

- A smaller top degree gives y = 0. Equal degrees give the ratio of the leading coefficients. A bigger top degree gives no horizontal asymptote.

7 Absolute extrema on a closed interval

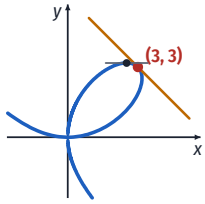


Label: both endpoints, the critical numbers inside, and the largest and smallest values.

1. f is continuous on [−2, 3], so both absolute extrema exist.
2. f'(x) = 6(x − 2)(x + 1) gives the critical numbers −1 and 2. Both are inside.
3. In the value table, the largest f is the max and the smallest is the min.

x	−2 (end)	−1	2	3 (end)
f(x)	1	**12, max**	**−15, min**	−4

8 Implicit curve and its tangent line

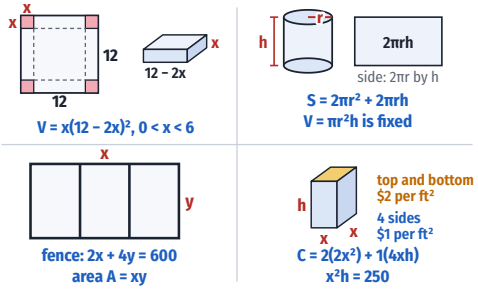


$x^3 + y^3 = 6xy$
$3x^2 + 3y^2y' = 6y + 6xy'$
$y'(3y^2 - 6x) = 6y - 3x^2$
$y' = (2y - x^2) / (y^2 - 2x)$
check: 27 + 27 = 6(3)(3) = 54
at (3, 3): y' = (6 − 9) / (9 − 6) = −1
tangent: y − 3 = −1(x − 3)
gray: horizontal tangent where $2y - x^2 = 0$ on the curve

Label: the point, a check that it is on the curve, the slope there, and the tangent line.

- A horizontal tangent needs the top of y' to be 0, and a vertical tangent needs the bottom to be 0, at a point on the curve.

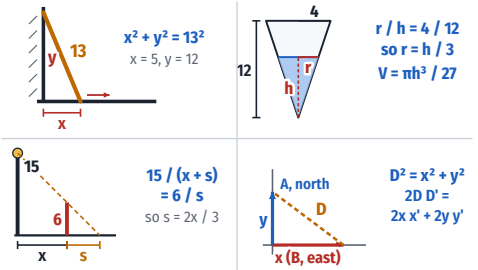
9 Optimization: four setups



Label: a letter on every length with units, the objective, the constraint, and the domain.

- Open box: V' = (12 − 2x)(12 − 6x) = 0 gives x = 2 in (0, 6), so V = 128 cubic inches.
- Can: the least material is at h = 2r. Pen: x = 150 ft and y = 75 ft. Box: x = 5 ft, h = 10 ft, and the cost is $300.

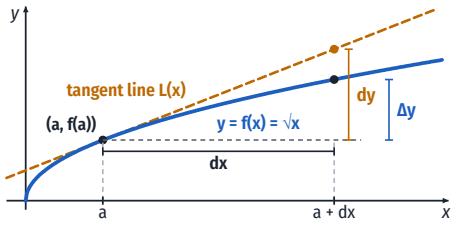
10 Related rates: four pictures



Label: changing lengths as letters, fixed lengths as numbers, and each rate with its sign.

- Cars at 60 and 80 mph: after 3 minutes y = 3 and x = 4, so D = 5 and D' = (4 · 80 + 3 · 60) ÷ 5 = 100 mph.
- Shadow: s = 2x/3, so the shadow grows at 2/3 of the walking speed.

11 Linear approximation: tangent line versus curve

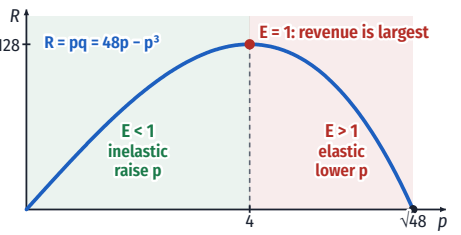


Label: the base point a, the run dx, dy up to the tangent line, and Δy up to the curve.

1. Pick a near x where f(a) and f'(a) are exact. Then $f(x) \approx f(a) + f'(a)\,dx$, with dx = x − a.

- dy is the rise along the tangent line and Δy is the true rise of the curve. They are close when dx is small.
- Concave down puts the tangent line above the curve, so the estimate is too big: $\sqrt{26} \approx 5 + (1/10)(1) = 5.1$ is a little high.

12 Elasticity and revenue



Label: the price where E = 1 at the top of R, the inelastic side, and the elastic side.

1. $q = 48 - p^2$ gives $E = 2p^2 \div (48 - p^2)$. E = 1 when $2p^2 = 48 - p^2$, so at p = 4.

- dR/dp = q(1 − E), so revenue rises with the price when E < 1 and falls when E > 1.

Math 16A Midterm 2 · Side 2 · Every rule and formula

Review sheet: memorize this, it is not allowed in the exam.

Derivative rules

Rule	Formula (any real n)	Example
Power	$(x^n)' = nx^{n-1}$	$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
Product	$(uv)' = u'v + uv'$	$(x^2e^x)' = 2xe^x + x^2e^x$
Quotient	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$	$\left(\frac{x}{x+1}\right)' = \frac{1}{(x+1)^2}$
Chain	$[f(g(x))]' = f'(g(x))g'(x)$	$[(3x+1)^5]' = 15(3x+1)^4$
Chain power	$(g^n)' = ng^{n-1}g'$	$(\sqrt{x^2+9})' = \frac{x}{\sqrt{x^2+9}}$

f	f'	With inside u	f	f'	With inside u
e^x	e^x	$e^u\,u'$	$\sin x$	$\cos x$	$\cos u \cdot u'$
a^x	$(\ln a)a^x$	$(\ln a)a^u\,u'$	$\cos x$	$-\sin x$	$-\sin u \cdot u'$
$\ln x$	$\frac{1}{x}$	$\frac{u'}{u}$	$\tan x$	$\sec^2 x$	$\sec^2 u \cdot u'$
$\log_a x$	$\frac{1}{(\ln a)x}$	$\frac{u'}{(\ln a)u}$	$\cot x$	$-\csc^2 x$	$-\csc^2 u \cdot u'$
$\ln\lvert x\rvert$	$\frac{1}{x}$	$\frac{u'}{u}$	$\sec x$	$\sec x \tan x$	$\sec u \tan u \cdot u'$
k	0	$(kf)' = kf'$	$\csc x$	$-\csc x \cot x$	$-\csc u \cot u \cdot u'$

The three functions whose names start with "co" (cosine, cotangent, cosecant) are the three with a minus sign.

- The power rule needs a constant exponent, as in x^5. The a^x rule needs a constant base, as in 5^x. For x^x, take ln of both sides first: the derivative is $x^x(\ln x + 1)$.
- $\sin^2 x$ means $(\sin x)^2$, so its derivative is $2 \sin x \cos x$. The derivative of $\sin(x^2)$ is $2x\cos(x^2)$.
- With three layers, write one factor per layer from the outside in: $\ln(\cos(e^{2x}))$ has derivative $[-\sin(e^{2x}) \cdot 2e^{2x}] \div \cos(e^{2x})$.

Exponents and logarithms

$\ln(xy) = \ln x + \ln y$, $\ln(x/y) = \ln x - \ln y$, $\ln(x^r) = r \ln x$, and $\log_a x = \ln x \div \ln a$.

- $\ln 1 = 0$, $\ln e = 1$, $e^{\ln x} = x$, $\ln(e^x) = x$, and $a^x = e^{x \ln a}$.
- $x^a x^b = x^{a+b}$, $x^a/x^b = x^{a-b}$, $(x^a)^b = x^{ab}$, $x^{-a} = 1/x^a$, and $x^{m/n}$ is the n-th root of x^m.
- $\ln x$ exists only for $x > 0$. A power of e is always positive. To solve $e^x = 5$, take ln of both sides: $x = \ln 5$.
- Simplify a log before you differentiate: $\ln(x^3 \div \sqrt{(x+1)}) = 3 \ln x - \frac{1}{2} \ln(x+1)$, whose derivative is $3/x - 1/(2(x+1))$.
- **TRAP** $\ln(x^2 + 5)$ cannot be split. Logs split only products, quotients, and powers.

Trig values and identities

Angle	0	$\frac{\pi}{6}$ (30°)	$\frac{\pi}{4}$ (45°)	$\frac{\pi}{3}$ (60°)	$\frac{\pi}{2}$ (90°)
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

- $\tan x = \sin x / \cos x$, $\cot x = \cos x / \sin x$, $\sec x = 1 / \cos x$, $\csc x = 1 / \sin x$, $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, and $\sin 2x = 2 \sin x \cos x$. Use radians: π radians is 180°.

For other angles, the first-quadrant reference angle gives the size and the quadrant gives the sign. All are positive in quadrant I, only sin in II, only tan in III, and only cos in IV.

- Example: $5\pi/6$ is in quadrant II with reference angle $\pi/6$, so its sin is $1/2$ and its cos is $-\sqrt{3}/2$.

L'Hospital's rule

If substituting gives $0/0$ or $\pm\infty/\pm\infty$, then the limit of f/g equals the limit of f'/g' (when that limit exists or is $\pm\infty$).

1. Substitute, and write the limit of the top and of the bottom. Name the form in words.
2. If the form is not $0/0$ or ∞/∞, stop: the substitution answer is the limit.
3. Differentiate the top and the bottom separately (no quotient rule), then substitute again. Repeat if needed.

- Rewrite $0 \cdot \infty$ as a quotient first, as in $x \ln x = (\ln x) \div (1/x)$. Combine $\infty - \infty$ into one fraction first.
- To know: $\lim_{x\to 0} \frac{\sin x}{x} = 1$, $\lim_{x\to 0} \frac{e^{3x}-1}{x} = 3$, $\lim_{x\to\infty} \frac{\ln x}{x} = 0$, $\lim_{x\to\infty} \frac{x^2}{e^x} = 0$, $\lim_{x\to 0^+} x \ln x = 0$.
- For large x, ln x grows more slowly than any power x^p with $p > 0$, and every power grows more slowly than e^x.
- **TRAP** $(x + \cos x)/(x + 2)$ as x goes to 0 has the form 1/2, which is not indeterminate, so the limit is 1/2.

Economic order quantity (lot size)

- M units a year come in batches of q. Each batch costs f to set up, each unit costs g, and storing one unit for a year costs k. The average inventory is half a batch.

$C(q) = fM/q + gM + kq/2$ for $0 < q \le M$. Then $C'(q) = -fM/q^2 + k/2 = 0$ gives $q = \sqrt{(2fM/k)}$.

- There are M/q batches a year, and g does not change the best q.
- Justify: $C''(q) = 2fM/q^3 > 0$ for $q > 0$, and q is the only critical number, so it gives the absolute minimum cost.
- **TRAP** If storage is charged on a full batch (rented space), rebuild C(q) from its parts with kq instead of kq/2.

Definitions to be able to state

A critical number of f is a number c in the domain of f where f'(c) = 0 or f'(c) does not exist.

- An inflection point is a point (c, f(c)) on the graph, with f continuous at c, where the concavity changes.
- Extreme value theorem: a function continuous on a closed interval [a, b] has an absolute max and an absolute min on [a, b].
- The tangent line at x = a is $y - f(a) = f'(a)(x - a)$. Velocity is v = s', acceleration is a = v' = s", and an object speeds up when v and a have the same sign.
- Marginal cost is C'(x), and the same for R and P. Revenue R = xp, profit P = R − C, and average cost is C(x)/x.

Sign chart and the two derivative tests

1. Find the domain. Write f' as one fraction and factor its top and bottom completely.
2. Critical numbers are numbers in the domain where the top or the bottom of f' is 0.
3. Put them and the numbers outside the domain on a line, and test one value inside each interval.
4. $f' > 0$ means increasing and $f' < 0$ means decreasing. Answer with open intervals, never joined across an excluded number.
5. Plus to minus is a relative max, minus to plus is a relative min, and no change is neither. Get f(c) from f.

Second derivative test

- At a c with $f'(c) = 0$, $f''(c) > 0$ means a cup, so a relative min, and $f''(c) < 0$ means a cap, so a relative max. If $f''(c) = 0$, or f'(c) does not exist, use the first derivative test.

Absolute extrema

Closed interval [a, b]: the closed interval method

1. Say f is continuous on [a, b], so the extreme value theorem says the absolute max and min exist.
2. Find the critical numbers, and throw out in writing any that are not in (a, b).
3. Evaluate f at the critical numbers left and at a and b. The largest value is the absolute max and the smallest is the absolute min. Give each value and the x where it happens.

Other intervals: the critical point theorem

4. If f is continuous on the interval and c is its only critical number there, a relative min at c is the absolute min, and a relative max at c is the absolute max.
5. For the other kind, look near the ends: if f(x) goes to ∞, there is no absolute max.

- Example: $x + 9/x$ on $(0, \infty)$ has the absolute min f(3) = 6 and no absolute max, because f(x) goes to ∞ as x approaches 0 from the right.

Optimization

1. Draw and label a picture. Define each variable in a sentence with units.
2. Write the objective as a formula and the constraint as an equation. Solve the constraint for one variable and substitute.
3. State the domain from the context, like $x > 0$ or $0 \le x \le 6$.
4. Find the critical numbers in the domain. Justify the max or min by the closed interval method (closed domain) or the critical point theorem (open domain).
5. Answer in a sentence with units, giving every dimension asked for.

- Example (card 9 box): $h = 250/x^2$ gives $C(x) = 4x^2 + 1000/x$, and $C'(x) = 8x - 1000/x^2 = 0$ gives x = 5 and h = 10. $C''(x) = 8 + 2000/x^3 > 0$ and 5 is the only critical number, so $300 is the least cost.

Elasticity of demand and revenue

$E = -(p/q)(dq/dp)$. In this course E is always positive.

1. Write q as a function of p, find dq/dp as a formula, and simplify E before putting in any price.
2. Put in the price and compare E with 1.

E	Demand is	To raise revenue
E < 1	inelastic	raise the price (they move together)
E = 1	unit elastic	nothing: revenue is at its max
E > 1	elastic	lower the price (they move oppositely)

- Why: $R = pq$, so $dR/dp = q + p(dq/dp) = q(1 - E)$, and $q > 0$. If demand gives p in terms of q, use $dq/dp = 1 \div (dp/dq)$.

To explain: "At p = 6, E = 6 > 1, so demand is elastic. Since $dR/dp = q(1 - E) < 0$, revenue falls when the price rises, so lower the price to raise revenue."

Implicit differentiation

1. Differentiate both sides with respect to x, treating y as a function of x: $(y^3)' = 3y^2y'$, $(e^y)' = e^y y'$, $(xy)' = y + xy'$.
2. Move the y' terms to one side and the rest to the other. Factor out y' and divide.
3. For a slope at a point, check the point is on the curve, then substitute both coordinates.

- Example: $x^2 + y^2 = 25$ gives $2x + 2yy' = 0$, so $y' = -x/y$. At (3, 4), which is on the circle since 9 + 16 = 25, the slope is $-3/4$.

Related rates

1. Draw a picture with letters on the changing lengths and numbers on the fixed ones. Name each variable with units.
2. Write the given and wanted rates as derivatives with respect to t. A shrinking quantity has a negative rate.
3. Write an equation that is true at every time, and differentiate both sides with respect to t. Only then substitute the numbers for the instant, finding missing values from the original equation.
4. Solve, and answer with units, saying whether the quantity is increasing or decreasing.

- Ladder, 13 ft, bottom pulled out at 2 ft/s, bottom 5 ft out: $2x(dx/dt) + 2y(dy/dt) = 0$ gives $2(5)(2) + 2(12)(dy/dt) = 0$, so $dy/dt = -5/6$. The top slides down at 5/6 ft/s.
- Cone tank 12 ft tall, top radius 4, draining 2 cubic ft per minute: $V = \pi h^3/27$, so $dV/dt = (\pi h^2/9)(dh/dt)$. At h = 6, $-2 = 4\pi(dh/dt)$, so the level falls at $1/(2\pi)$ ft per minute.

Linear approximation and error

$L(x) = f(a) + f'(a)(x - a)$. The differential $dy = f'(x)\,dx$ approximates the true change $\Delta y = f(x + dx) - f(x)$.

- Error in Q from a measured x with maximum error dx: $dQ = Q'(x)\,dx$, at the measured x, in the units of Q.

Relative error = dQ/Q. Percentage error = 100 × dQ/Q. Neither has units.

- If you are given a diameter but the formula uses the radius, halve both the value and the error.
- Example: r = 6 cm with error 0.02 cm gives $dV = 4\pi r^2 dr = 4\pi(36)(0.02) = 2.88\pi$ cubic cm. Since $V = 288\pi$, the relative error is 0.01, or 1%.

Geometry formulas

Shape	Area or volume	Perimeter or surface area
Rectangle	lw	2l + 2w
Triangle	½bh	add the three sides
Circle	πr^2	circumference $2\pi r$
Square base box	x^2h	closed $2x^2 + 4xh$, open $x^2 + 4xh$
Cylinder	$\pi r^2 h$	closed $2\pi r^2 + 2\pi rh$, side $2\pi rh$
Cone	$(1/3)\pi r^2 h$	side $\pi r\sqrt{(r^2 + h^2)}$
Sphere	$(4/3)\pi r^3$	$4\pi r^2$

- Right triangle: $a^2 + b^2 = c^2$. To minimize a distance, minimize its square.

Sentences that earn the justification points

- ☐ "f has a relative max at x = c because f' changes from positive to negative at c."
- ☐ "f'(0) = 0, but f" does not change sign at 0, so there is no inflection point at x = 0."
- ☐ "f is continuous on the closed interval [a, b], so by the extreme value theorem it has an absolute max and min."
- ☐ "The critical number x = 5 is not in [0, 4], so I discard it."
- ☐ "x = c is the only critical number in (0, ∞) and C"(c) > 0, so by the critical point theorem it gives the absolute minimum."

Most common mistakes

- **TRAP** Forgetting the inside derivative. The derivative of e^{-2x} is $-2e^{-2x}$.
- **TRAP** (uv)' is never u'v', and the quotient rule top is u'v − uv' in that order. The power rule does not apply to 2^x, whose derivative is $(\ln 2)2^x$.
- **TRAP** The y-value of a point always comes from f, never from f' or f".
- **TRAP** f'(c) = 0 is not always an extremum, and f"(c) = 0 is not always an inflection point. Show the sign change.
- **TRAP** In optimization, "f"(c) > 0" alone proves only a relative min. Add that c is the only critical number.
- **TRAP** Substituting the instant's numbers before differentiating turns a changing quantity into a constant.
- **TRAP** (xy)' = y + xy', not y'. A negative E means a dropped minus sign.
- **TRAP** dy is only the change. The new value is f(a) + dy.